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SUPERSEDING
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MILITARY SPECIFICATION

TUNGSTEN POWDER

1. SCOPE

1.1 Scope.- This specification covers one class of tungsten powder.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-M-151 - Metals, General Specification for inspection of

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical.-The chemical composition shall meet the requirements shown in table I.

TABLE I - Chemical requirements

Tungsten, percent, minimum	99.00
Oxygen, percent maximum	.50
Iron, percent, maximum	.10
Non Volatile Residue, percent, maximum	.10

3.2 Density.-The density of tungsten powder shall be not less than 35 nor more than 65 g. per cubic inch as determined by Scott Volumeter.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Lot.-A lot shall consist of not more than 10,000 pounds nor less than 1,000 pounds of tungsten powder produced by the same process and submitted for inspection at one time.

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4.2 Sampling.-A sample of approximately 200 g. shall be selected by the inspector from each lot for chemical analysis and powder density.

4.3 Density.-Assemble a Scott Volumeter, having a suitable screen in the upper hopper, so that the tared receiving cube, one cubic inch in volume, is directly under the lower funnel and rests on the base of the apparatus. Slowly pour the sample into the hopper and brush, if necessary, through the screen in order to fill the receiving cube to overflowing. By means of a spatula, carefully strike off the excess powder from the cube, taking care not to jar the cube during this operation. Weigh the cube and contents, and from the volume and weight of powder calculate its apparent density. Make three such tests and calculate the average of the apparent density values.

4.4 Composition.-The following methods of analyses shall be used unless otherwise approved.

4.4.1 Non-volatile residue.

4.4.1.1 Apparatus.

a. Source of dry HCl gas - preferably a cylinder of compressed gas with special adapter and needle valve.

b. Calibrated flowmeter (liquid indicator - light white mineral oil, U. S. P. or equal).

c. Combustion furnace; hinged type, with pyrometer and rheostat.

d. Combustion tube, silica, with reduced exit end for attachment of rubber tubing. Latter leading to hood.

e. Combustion boats. - Coors porcelain #8 or equivalent.

4.4.1.2 Procedure.-Ignite two porcelain boats over a free flame to constant weight (± 0.2 mg). Place duplicate weighed samples of approximately 2 g. each in the tared porcelain boats and ignite over a free flame for 1 hour. Place the ignited samples in a furnace kept at 600°C and pass dry HCl gas through for one hour at a rate of flow of 30 to 50 ml of HCl per minute. Place boats in a desiccator and cool for 30 minutes. Weigh. Replace the boat and contents in the furnace and repeat the procedure described above until a constant weight is attained. Calculate the percent non-volatile residue as follows:

$$\text{Percent non-volatile residue} = \frac{100A}{W}$$

Where A = gain in weight of each boat
W = weight of sample.

4.4.2 Iron

4.4.2.1 Procedure - Place 10.00 g. of the tungsten powder in a 100 ml platinum dish and add approximately 25 ml of concentrated HF. Pour about 20 ml of concentrated HNO_3 into a beaker. Using a dropper, add this HNO_3 to the sample until all the metal is dissolved, and then pour in the rest of the acid. Evaporate the solution to dryness. Cool and wet the residue with distilled water. Cover the tungsten oxide with a considerable excess of granular anhydrous Na_2CO_3 and slowly heat the dish until water is expelled. Bring the contents to fusion at red heat and rotate the liquid until all parts of the inner surface of the dish have been in contact with the liquid. Add about 60 ml of water to the cooled fusion and warm until the melt loosens. Transfer the liquid and undissolved portion of the melt to a 250 ml beaker and wash the dish thoroughly, adding the washings to the remainder of the melt. Heat the solution just below the boiling point until all soluble salts dissolve. Pass H_2S gas into the solution at room temperature for 10 to 15 minutes. Digest at 60°C for 1/2 hour. Filter through

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a Whatman No. 40 paper or equivalent and wash as free from tungsten as possible with a hot 1 percent Na_2CO_3 wash solution which has been treated with H_2S for 2 to 3 minutes. Discard the filtrate. Add about 10 ml of concentrated HCl to the platinum dish and digest and manipulate the liquid until all the Fe_2O_3 is dissolved. Dilute the solution to about 25 ml and heat to incipient boiling. Carefully pour the dilute HCl through the filter containing the Fe_2O_3 + FeS, catching the solution in a clean 250 ml beaker. Wash the filter thoroughly with water and ignite in a 40 ml porcelain crucible. Cool. Add 10 ml of concentrated HCl and heat until all the Fe_2O_3 is dissolved. Add 10 ml of hot water, mix and filter through a Whatman No. 40 paper or equivalent, collecting the filtrate in the 250 ml beaker containing the remainder of the Fe solution. Wash well with hot 1:10 HCl. Add approximately 1 ml of concentrated HNO_3 to the filtrate and digest for several minutes. Precipitate the iron with NH_4OH and filter on No. 41 paper with pulp. Ignite and weigh as Fe_2O_3 and calculate to Fe

$$\text{Percent Fe} = \frac{69.9A}{W}$$

where A = weight of residue
W = weight of sample.

4.4.3 Oxygen

4.4.3.1 Apparatus

4.4.3.1.1 Source of H_2 gas - preferably a cylinder of compressed gas.

4.4.3.1.2 Purification train (in the order listed).

Safety bottle (gas washing bottle reversed).
Gas washing bottle containing 100 g. KOH + 7 g. pyrogallic acid + 100 ml H_2O.
Safety bottle
Gas washing bottle containing concentrated H_2SO_4.
U tube containing ascarite.
U tube containing anhydrous $Mg(ClO_4)_2$.

4.4.3.1.3 Combustion furnace, tube and boats - as in determination of nonvolatile residue. The excess H_2 is led into a hood (Caution - no flames).

4.4.3.2 Procedure - Ignite two porcelain boats over a free flame to constant weight (+.2 mg). Place duplicate weighed samples of approximately 20 g. each in the tared porcelain boats, transfer to a cold furnace, and pass H_2 gas through at a slow rate. Raise the furnace temperature to 900-950°C and maintain for 1 hour. Switch off the furnace and without disconnecting any portion of the train, remove the combustion tube from the furnace and place on a stand. Cool the tube by means of an electric fan until comfortable to the touch. Stop the flow of H_2 and transfer the boats to a desiccator. Cool for 30 minutes and weigh. Repeat the above procedure, with the omission of ignition over a free flame, to constant weight, heating for 1/2-hour periods. Calculate the percentage of oxygen as follows:

$$\text{Percent oxygen} = \frac{100A}{W}$$

where A = loss of weight of each sample
W = weight of sample.

4.4.4 Tungsten

4.4.4.1 Preparations of solutions

4.4.4.1.1 Cinchonine solution - Dissolve 125 g. of cinchonine in 1 liter of 1:1 HCl.

4.4.4.1.2 Cinchonine wash solution - Dilute 30 ml of the cinchonine solution (par. 4.4.4.1.1) to 1 liter with water.

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4.4.4.2 Procedure - Accurately weigh a previously dried portion of approximately 1 g. of the sample and transfer to a 60 ml covered platinum crucible or dish. Add 5 ml of HF and then HNO_3 drop by drop until the sample is dissolved. Remove the cover and rinse with water. If necessary scrub with a policeman and wash down with water. Add 15 ml of 1:1 H_2SO_4, cover with a raised watch glass and heat cautiously on a sand bath to dense white fumes. Allow to cool. Again scrub the cover with a policeman and wash down with water. Transfer as much of the residue in the dish as possible to a 600 ml beaker with water. Scrub the dish and transfer the remainder of the contents to the beaker. Rinse the dish with a little warm 1:1 NH_4OH followed by water and then a few ml of hot 1:1 HCl. Add the rinsings to the beaker. Repeat the treatment starting with rinsing the dish with 1:1 NH_4OH.

Dilute the contents of the beaker to about 150 ml with water, add 10 ml of HCl and boil for 5 minutes. Remove from the source of heat and dilute to 450 ml with water. Add 10 ml of cinchonine solution and some ashless paper pulp and allow to stand overnight. Filter on a 11 cm Whatman No. 40 paper or equal, adding a little ashless paper pulp. Wash thoroughly with cold cinchonine wash solution, and finally several times with warm 1:99 HCl. Place the paper and residue in a weighed platinum crucible, and heat gently in a muffle furnace at 300°-400°C until the moisture is expelled and the paper carbonized. Raise the temperature to about 600°C and ignite until the carbon is consumed. Allow to cool. Add a few drops of HNO_3 and several ml of HF. Heat to dryness in an air bath. Then ignite in a muffle furnace for 30 minutes at not over 750°C. Cool in a desiccator and weigh and repeat to constant weight ± 0.5 mg.

To the ignited residue add 5 g. of anhydrous Na_2CO_3 and mix thoroughly. Fuse, running the fusion around the sides of the crucible to remove all WO_3. Dissolve the fusion in hot water, add a little alcohol, heat to boiling, and filter on Whatman No. 40 paper or equivalent. Wash the crucible and residue thoroughly with hot water. Place the residue and paper in the crucible and ignite to burn off paper and remove carbon. Then add 2 g. of anhydrous Na_2CO_3 to the residue and fuse again. Repeat the solution, filtration, and washing of the fusion as before. The washing shall be as thorough as possible to remove the last traces of Na_2CO_3. Transfer the residue to a weighed platinum crucible and ignite as before. Cool and weigh. Run a blank on the same weight of Na_2CO_3 as was used with sample following the same procedure to determine the amount of insoluble residue introduced from this source. Calculate the percentage of tungsten as follows:

$$\text{Percent tungsten} = \frac{79.31\ (A-(B-C))}{W}$$

where A = grams of impure WO_3
B = grams of insoluble residue from Na_2CO_3 fusion
C = grams of insoluble residue in Na_2CO_3 (blank)
W = grams of sample.

5. PREPARATION FOR DELIVERY

5.1 Packing.-Unless otherwise specified, tungsten powder shall be packed dry in airtight moistureproof metal containers having a maximum capacity of 15 gallons so constructed as to insure acceptance by common or other carrier for safe transportation at the lowest rate.

5.2 Marking.-Containers for shipments direct to Government installations shall be marked in accordance with Military Standard MIL-STD-129.

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6. NOTES

6.1 Intended use.-The material covered by this specification is intended for use in the manufacture of tungsten carbide cores for hyper-velocity armor-piercing projectiles.

6.2 Ordering data.-Purchasing documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity.

c. If packaging other than provided by 5.1 is required.

6.3 This specification replaces U. S. Army Specification No. 57-26-1.

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